The Association Between Vitreous Floaters and Psychological Distress in Young Adults (Ages 16 to 26)

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ABSTRACT

Vitreous floaters are often dismissed as benign, age-related visual phenomena. However, emerging research suggests they may negatively impact mental health and quality of life. Most existing studies focus on older populations, limiting the applicability of findings to younger individuals. This study investigates the correlation between vitreous floater severity and psychological distress in young adults (ages 16–26). A 37-question online survey, incorporating validated instruments such as the Perceived Stress Scale (PSS), State-Trait Anxiety Inventory (STAI), and the Vitreous Floaters Symptom Questionnaire, was distributed globally via Reddit forums. Sixty-eight participants (11 controls, 57 symptomatic) completed the survey. Nonparametric statistical analysis revealed no significant associations between floater severity and levels of stress or anxiety. However, participants with more severe floaters were more likely to report recent symptom onset. While clear correlations were not observed, the findings suggest that clinicians might consider incorporating mental health assessments and offering non-surgical, therapeutic interventions to better support affected young adults.

Keywords: vitreous floaters, psychological distress, young adults, survey, therapeutic approaches

1. INTRODUCTION

1.1. Overview of vitreous floaters. Vitreous floaters, or more commonly known as eye floaters, a scientifically named myodesopsias. They are described by patients as squiggly lines, spots, spider-like shapes, zigzags, or dots in their field of vision. They are more visible when viewed against bright, white backgrounds due to the dark shadows created by the vitreous opacities being accentuated (Skowronek & Święch, 2023). Floaters are scattered clumps of collagen fibers that move in the jelly-like substance (vitreous) in the eye and cast shadows on the macula of the retina. The retina is the part of the eye responsible for sight and translates the signals of the perceived fibers as vitreous floaters. Floaters are perceived when light scatters around these structures in the vitreous body or at the posterior vitreous cortex. While vitreous floaters can shift around in the vitreous and appear to go away due to viscous drag, the condition is permanent (Cleveland Clinic, 2023). In a survey of 603 self-reporting participants, 76% of participants reported seeing vitreous floaters and 33% reported vision impairment due to them (Webb et al., 2013).

1.2. Causes. Vitreous floaters are caused by molecular changes within the vitreous body and alter the vitreous structure. As humans age, the internal structure of the vitreous body changes and vitreoretinal adhesion weakens (Milston et al., 2016). At human birth, the vitreous appears as a solid and clear gel filling in the eye. The transparency of it allows unhindered transmission of light to the retina for photoreception to be perceived (Sebag, 2011). However, with aging, vitreous gel liquefaction occurs, being first evident at age four. The condition arises when hyaluronan separates from collagen within the vitreous body, leading to the clumping and cross-linking of collagen fibrils into larger fibers that scatter incoming light. (Milston et al., 2016). Besides aging, floaters typically occur due to posterior vitreous detachment (PVD), when the vitreous or vitreous humor lifts up from the surface and pulls on the retina, creating tension (Cleveland Clinic, 2023). PVD is believed to be a salubrious result of evolution, preventing more harmful conditions from occurring (Sebag, 2011). Another less common and more damaging causes are due to retinal tears and retinal detachment, which are due to the pulling of the vitreous, causing a breakage in the retina (Cleveland Clinic, 2023).

1.3. Risk. The risk of PVD-caused eye floaters increases with age due to the progression of collagen breakdown and the shrinkage of the vitreous. PVD affects 60% of people over the age of 70 and 50% of PVD cases occur in people aged 50 years and older (Skowronek & Święch, 2023). Older women are at greater risk of PVD potentially due to the modified biochemical composition of vitreous from hormonal changes associated with menopause (Milston et al., 2016). Additionally, myopia (nearsightedness) increases risk of PVD which is attributed to earlier and accelerated vitreous liquefaction due to a longer axial length (Hayashi et al., 2020). One study found that myopes were three and a half times more likely than emmetropes to report greater floater severity (Webb et al., 2013). Furthermore, vitreoretinal disorders such as Marfan, Stickler syndrome. Wanger disease is associated with greater floater risk (Milston et al., 2016). Having a history of other pathological conditions such as uveitis, vitreous hemorrhage, and ocular trauma increases the risk of floaters (Skowronek & Święch, 2023).

1.4. Current Treatments. Ophthalmologists commonly confirm vitreous floaters through ophthalmoscopy or biomicroscopy. Other detection methods for finer floaters include ultrasound, spectral domain OCT, scanning laser ophthalmoscopy, dynamic light scattering (Milston et al., 2016). Patients are recommended to avoid seeking surgery as treatment for vitreous floaters unless they are numerous and cause severe visual disturbances. Additionally, patients are told that they will adapt to the floaters and they will settle inferior to the visual axis settling because of neuro-psychological adaptation (Cleveland Clinic, 2023). However, a study of 266 patients found acute and chronic floaters (duration) did not improve their utility values, indicating that patients do not adapt to their symptoms (Wagle et al., 2011). There are two common types of treatment procedures for more severe floaters: floaterectomy by vitrectomy and floaterrhexis, by Nd:YAG laser vitreolysis. Vitrectomy permanently removes vitreous opacities while Nd:YAG laser breaks large floaters into smaller ones and dissolves them. Vitrectomy has evidence suggesting its effectiveness, including several studies that reported improvements in acuity ranging from 26% to 50% following the procedure and survey results showing 85% to 100% of participants being satisfied with their treatment. However, both of these invasive treatments have risks including cataract formation, retinal tears and detachment, intraocular pressure problems, vitreous hemorrhage, endophthalmitis and chronic glaucoma (Milston et al., 2016). There is also no guarantee that these procedures will remove all floaters from the eye. Furthermore, some studies have suggested that supplementation with anti-glycation and antioxidant substances may reduce visual complaints associated with vitreous opacities. For example, Inositol has been shown to inhibit collagen glycoxidation, reducing the formation of collagen cross-links and helping to preserve its structural integrity. Additionally, L-lysine has demonstrated benefits in preventing collagen glycation and is beneficial in treating eye conditions, particularly those affecting the vitreous body. However, effectiveness of these treatments still needs to be demonstrated (Skowronek & Święch, 2023).

1.5. Impact on quality of life. There is numerous evidence linking vitreous floaters with poor quality of life. Floaters cause contrast sensitivity (CS) to diminish and straylight to increase, thereby making people unhappy with their vision. The deterioration of vision caused by the floaters presents visual symptoms including hazy vision, difficulty driving at night, and decreased facial recognition (Milston et al., 2016). In a study by Webb et al., over two thirds of floater patients struggled with reading and night driving due to their floaters (Webb et al., 2013). Mason et al. support this claim, finding that 50% of patients have struggles reading and 30% experience difficulties driving due to symptomatic vitreous floaters (Mason et al., 2014). In a study by Zou et al., they found that floaters can diminish their patient's perceived quality of life (Zou et al., 2012). Similar decreasing quality of life has been found in general impairment. In a large cohort study, they found that mild and severe visual impairment was associated with a deterioration in the patient's quality of life (Taipale et al., 2019). Additionally, a study by Wagle et al. revealed that the utility values for floaters are lower than those for mild angina, mild stroke, colon cancer, and asymptomatic HIV infection, indicating that floaters present a significant impact on quality of life as compared to other health-related conditions. Moreover, in one study, patients under fifty-five reported that they are willing to risk a 11% risk of death and accept a 7% risk of blindness to have their floaters completely removed. They are also willing to trade 1.1 years out of every 10 years to be rid of their symptomatic floaters (Wagle et al., 2011).

This study seeks to answer the research: To what extent is vitreous floater severity correlated with psychological distress in young adults (ages 16-26).

2. LITERATURE REVIEW

2.1. Influence of floaters on psychological distress. Throughout several studies, vitreous floaters have been associated with increased psychological distress, such as anxiety, depression, and stress. A meta-analysis finds across five studies that people with floaters showed significantly higher levels of depression, perceived stress, state anxiety, and trait anxiety compared to the control group (Senra et al., 2022). Another study supports these findings as higher anxiety and depression scores were significantly associated with floater patients within a sample of ninety patients (Gouliopoulos et al., 2024). A study by Kim et al. adds to these results, finding that the severe discomfort group showed higher depression, perceived stress, and state and trait anxiety compared to the other two milder symptom groups. This result shows that the severity of the floaters is correlated with greater psychological distress. However, there is a possibility that the patients with greater depression ranked and perceived their floater symptoms as being more severe (Kim et al., 2017). Additionally, study by Senra et al. on a UK sample finds negative personality traits are associated with eye floater patients, such as an increased neuroticism score and a reduced score in extraversion. Patients experiencing severe floater symptoms have been found to have higher underlying levels of neuroticism, which can explain their worries and effects on quality of life despite a benign non-sight threatening condition (Senra et al., 2024). A study on dry eye disease has also found higher levels of neuroticism in patients (Tananuvat et al., 2022). Floater symptoms are linked to more than just reduced visual acuity. A study found that individuals with SVF were two and a half times more likely to experience depressive states than healthy controls without floater symptoms, even though both groups had similar levels of visual acuity (Wu et al., 2020). Similarly, psychological distress has been associated with other eye diseases. One meta-analysis found that the prevalence of depression among macular degeneration, dry eye disease, glaucoma and cataract have a pooled 25% prevalence of depression (Zheng et al., 2017). Additionally, depression has been found in one-third of older adults with visual impairment, which is twice the prevalence rate found in those without visual impairment (Tan et al., 2020).

Senra et al. offer an explanation to the possible connection between floaters and psychological distress. They suggest that psychological symptoms are not directly associated with floaters, but due to other psychological factors such as perceived social support, anticipatory anxiety of future blindness or potential underlying conditions. This hypothesis has been claimed in other ophthalmic conditions such as age-related macular degeneration and diabetic retinopathy. Scholars suggest that the correlation between distress and ocular concerns lies in “psychological factors such as anticipatory anxiety of going blind in the future, fear of intravitreal injections, uncertainty about the prognosis, poor self-esteem, and poor perceived social support" (Senra et al., 2022). Similarly, a study on dry eye disease also finds that depression is not directly related to the symptoms of the disease, rather "psychological factors such as anxiety sensitivity, health anxiety, poor resilience and subjective well-being, and personality traits such as neuroticism and isolation tendencies" (Yilmaz et al., 2015).

2.2. Treatment of psychological distress. Various treatments have been proven to alleviate psychological distress associated with floaters. One study that analyzed the effectiveness of vitrectomy in 45 patients found improvements to both vision and mental health post-surgery (Goh et al., 2022). A study by Wu et al. found similar results, concluding that vitrectomy can effectively relieve the depression associated with the symptomatic floaters (Wu et al., 2020). Another study reports a significant reduction in anxiety and improvement in vision among patients who underwent Nd: Ultra-Q Reflex laser vitreolysis for floater vaporization (García et al., 2021). However, Senra et al. suggest that these quality of life improvements may be a placebo effect due to patients self-reporting and fearing the floaters (Senra et al., 2022).

Floater experience is unique to every patient and is based on their "perception of the disease, the personal explanation, the solutions tried, the trust placed in medicine, self-construction, and the dispersion of dependency" (Cipolletta et al., 2012). Some patients are more susceptible to emotional distress due to their personality traits and history compared to patients who find them to be an inconvenience. By recognizing the traits of patients who are more likely to experience a reduced quality of life, clinicians manage the patients' expectations and offer them non-surgical therapeutic approaches to manage their symptoms (Senra et al., 2024). Clinicians can support patients by offering them better explanations and providing therapy to ease their symptoms. Since mental health conditions are treatable, early identification of mental disorders of patients who are at risk can seek treatment to improve their overall well-being (Parravano et al., 2021).

3. GAP ANALYSIS

Younger patients typically experience vitreous floater symptoms associated with Cloquet's canal, leading them to appear in the inferotemporal area of the visual field. In contrast, elderly individuals experience floaters due to vitreous liquefaction, where floaters are found at the margin of a lacuna. Additionally, older patients typically experienced a single floater when it was related primarily to prepapillary opacities while they experienced numerous floaters due to vitreous hemorrhage. This difference in floater location is potential evidence that floater type, formation and experiences differ between age groups. As previously discussed, elderly are at greater risk of PVD. In one study, the number of symptomatic eye floaters without PVD was significantly higher in patients under fifty. Because patients with no PVD also tend to observe a single or few floaters, the perception of the floaters will differ between age groups (Murakami et al., 1983).

Generally, mental health differs between younger and older age groups. One study finds that "anxiety and depression decrease between 20 and 64 due to work influences and negative social relationships with family and friends" (Jorm et al., 2005). Additionally, results show that during COVID, older people reported less "pandemic-related stress, less life change, less social isolation, and lower negative relationship quality than younger people" (Birditt et al., 2021). Another source adds the psychological differences to this conclusion, revealing that older adults have "lower salivary cortisol levels in the stress and control conditions and lower stress-induced cortisol increase." They also have lower heart rate responses and lower physiological reactivity, suggesting that the two groups will react differently to floater symptoms (Mikneviciute et al., 2023).

A study suggests that younger people may react differently to floaters due to variations in personality traits. In their UK sample, they reported that age is positively correlated with conscientiousness and agreeableness and negatively correlated with extraversion, neuroticism and openness (Senra et al., 2024). These differing traits potentially reveal how responses to floaters may differ between age ranges, with older adults responding more positively. Patients have reported not noticing floaters as an issue when they accept it. Due to older adults having more positive personality traits and not needing to live with floaters for most of their lives, they are more likely to accept the floaters, thereby not suffering from as many psychological symptoms (Cipolletta et al., 2012).

One study on the effectiveness of a treatment finds that younger patients with vitreous floaters had higher levels of depression compared to older individuals. They attribute this trend due to the demand for good visual quality, as they tend to spend longer periods reading, driving, working with digital products. However, this study does not consider other psychological distress factors, such as anxiety and stress (Wu et al., 2020).

Studies such as by Kim et al., Wu et al., Parravano et al., and Senra et al., have failed to include large samples of younger populations, making their conclusions not as relevant to this demographic if their condition differs. For example, a systematic review of five studies have a combined patient age range of 32 to 78 years old (Senra et al., 2022). This exclusion of a younger age range makes the paper ineffective at drawing conclusions that relate to all ages.

Considering the difference in floater location and formation, stress response and personality differences between younger and older populations, this paper aims to address the gap of research on the younger population (ages 16 to 26) to determine the psychological distress trends associated with vitreous floaters.

4. METHODOLOGY

4.1. Overview. This study aims to collect quantitative data from distributed surveys. Survey allowed for a wide population to efficiently be reached and measurable, comparable data to be collected. Distributing the survey online made it easy to gather responses from individuals around the world within the target age range. The questions focus on collecting information on vitreous severity and psychological distress, including both stress and anxiety. Additionally, the different severities will be compared amongst controls.

*4.2. Participation. S*urveys were distributed among various Reddit channels. Voluntary sampling, where participants self-selected to be part of the research, was used to collect participant data. This sampling outreach allows for gathering diverse perspectives and experiences from people all over the world. On January 15, 2025, surveys were distributed on various Reddit channels (Table 1). Channels were chosen based on relevance to the study's target demographic, symptom focus, and survey participation likelihood to ensure a diverse sample size. The survey was posted on each channel twice; on January 15th and February 1st. Between January 15, 2025 and February 15, 2025, the survey remained open. Reddit channels were chosen based on the target audience.

4.3. Survey Design. The survey comprises thirty-nine questions, requiring participants to rank prompts on a Likert Scale. The survey questions were modeled after previous research, primarily Kim et al. and Senra et al. Definitions of terminology and images of vitreous floaters were provided to adequately inform participants. To collect data on the severity and characteristics of vitreous floater symptoms, the Vitreous Floaters Symptom Questionnaire adopted from van Overdam et al was used. This questionnaire consists of seven questions. Perceived stress levels were evaluated using the Perceived Stress Scale (PSS) adopted from Cohen et al.. It consists of 10 questions which participants are asked to rate on a five-point Likert scale, ranked 0 to 4. Question points are summed to yield a final score, where higher scores indicate higher perceived stress levels. Trait anxiety was measured using the State-Trait Anxiety Inventory (STAI) adopted from Spielberger et al.. The STAI consists of 20 questions for state anxiety and 20 questions for trait anxiety, all rated on a five-point Likert scale and ranked 0 to 4. Trait anxiety questions were omitted due to this study focusing on short-term anxiety levels and to reduce the risk of survey fatigue.

4.4. Ethicality. Factors were taken into account to ensure this study is ethical. Participants must read an Informed Consent Form and sign it with the date, thereby giving implied consent when they submit the survey.

All information collected by the survey is anonymous, as participants were not required to submit sufficient information that may reveal their identities. While participants did have the option to submit their emails to receive a copy of the study's findings, their emails are not associated with their answers. Data was secured safely and was only viewed by the Principal Investigator. This study has also undergone an Internal Review Board process and been accepted by the X Internal Review Board.

4.5. Statistical analysis. The program JASP was utilized to perform statistical analyses on the data. Statistical significance was defined as $p<0.05$, while borderline significance was considered when $p \geq 0.05$ and $p<0.08$. To assess the distribution of each variable, a Shapiro-Wilk test was conducted (see Table 2). Only the Floater group ($p=.007$) and the Mild Anxiety STAI group ($p<.001$) met the threshold to assume normality. Additionally, since each subgroup included fewer than 30 participants, the Central Limit Theorem could not be applied to justify normality. As the remaining variables did not follow a normal distribution, nonparametric tests were employed. The Mann-Whitney U test was used to compare floater and non-floater groups. To assess differences among floater severity levels, categorized as mild, moderate and severe, the Kruskal-Wallis test was employed, followed by a post hoc Dunn test. These nonparametric tests were selected as they do not assume normality and are well-suited for analyzing small sample sizes. The comparisons focused on perceived stress (PSS scores), trait anxiety (STAI scores), and subjective floater characteristics.

Table 2

Shapiro-Wilk test p-values for assessing normality of categorical variables

	Floaters	Controls	PSS Score			STAI Score		
			Mild	Moderate	Severe	Mild	Moderate	Severe
P-value of Shapiro-Wilk	0.007	0.472	0.787	0.802	0.603	<0.001	0.824	0.728

4.6. Replicability. This study is replicable because the survey instruments are publicly available and validated. I've also documented the exact Reddit channels used for distribution, and the same statistical methods I applied can be repeated. That said, results might vary depending on the participant sample collected in future replications.

5. RESULTS

In this study, 68 participants were included: 57 individuals with vitreous floaters and 11 controls without vitreous floaters. The average age of the study was 21.6, with the lowest participant age being 16 years old and the highest 26 years old (Table 3).

The PSS and STAI scores were compared between the floater and control groups using a Mann-Whitney U nonparametric test (Table 4). No significant difference was found between the two groups for either score (PSS: p=.253; STAI: p=.330). Observing the means, the floater group showed elevated anxiety levels, with a difference in PSS scores of 1.92 (M=24.19) compared to the control group (M=22.27). Similarly, the floater group showed elevated trait stress levels, with a difference in STAI scores of 3.563 (M=51.018) compared to the control group (M=47.455).

Table 3

Descriptive statistics of participant age by control status and floater severity

	Age				
			Floater Severity		
	Control	Floaters	Mild	Moderate	Severe
Valid	11	68	25	16	16
Mean	19.455	21.559	21.600	21.438	23.063
Standard Deviation	2.841	3.234	3.317	3.386	2.568
Minimum	17.000	16.000	16.000	16.000	18.000
Maximum	26.000	26.000	26.000	26.000	26.000

Table 4

Comparison of psychological distress across floater and control groups

	Group	N	Mean	p-value[a]
PSS Score	Control	11	22.273	0.253
	Floaters	57	24.193	
STAI Score	Control	11	47.455	0.330
	Floaters	57	51.018	

[a]Mann-Whitney U non-parametric test

Floater individuals were divided according to their degree of floater discomfort that they perceived. There were 25 participants that reported their floater discomfort as "mild," 16 as "moderate," and 16 as "severe." Floater symptoms were compared based on their degree of discomfort (Table 5). Most patients (mild: 100%, moderate: 94%, severe: 94%) described that their floater discomfort started several months or years ago. The trend is that patients with more severe discomfort experienced the onset of their discomfort more recently. Mild discomfort participants statistically and borderline statistically experienced their floaters longer than both severe ($p=.027$) and mild ($p=.051$) participants. As the severity of discomfort magnified, the daily frequency of discomfort increased. Between mild and moderate discomfort and mild and severe discomfort individuals, the frequency of the floaters is significant (for both, $p<.001$). Mild discomfort individuals were significantly more likely to report their floater "getting better" or "same" since its initial onset compared to severe discomfort individuals ($p<.001$). There were no obvious trends in the daily frequency of photopsia between any of the floater severity groups.

PSS scores were compared against floater severity to test for differences (Table 6). According to the Krushkal-Wallis Test, there was no statistical difference in PSS scores across floater severity groups and control groups ($p=0.218$). However, when observing Dunn's Post Hoc Comparison, the control and severe groups ($p=0.072$) and moderate and severe groups ($p=0.078$) were borderline statistically significant. These differences suggest that perceived stress levels slightly differed with floater severity. The descriptive statistics (Table 7) show a trend where mean PSS scores increase with severity, with control groups having the lowest scores ($M=22.273$), then mild ($M=23.080$), moderate ($M=23.375$), and severe the highest ($M=26.750$).

STAI scores were compared against floater severity to test for differences (Table 6). The Krushkal-Wallis Test did not reveal any statistical differences in STAI scores across floater severity groups and control groups ($p=0.495$). Dunn's post hoc comparisons revealed no statistically significant differences between any of the groups, as all p-values exceeded 0.167. The descriptive statistics (Table 7) show a trend where mean STAI scores increase with severity, with control groups having the lowest scores ($M=47.455$), then mild ($M=48.640$), moderate ($M=51.813$), and severe the highest ($M=53.938$). However, the means are not sufficient evidence to suggest a correlation between floater severity and psychological distress.

Table 5

Comparison of vitreous floater symptom characteristics based on reported levels of discomfort

	Mild (M) discomfort (n = 25)	Moderate (Mo) discomfort (n = 16)	Severe (S) discomfort (n = 16)	p-values[a]		
				M - Mo	M - S	Mo - S
Onset of discomfort, *n* (%)				0.051	0.027	0.728
One or two days ago	0 (0)	0 (0)	0 (0)			
Within a week	0 (0)	0 (0)	0 (0)			
Within a month	0 (0)	1 (6)	1 (6)			
Several months ago	4 (16)	7 (44)	8 (50)			
Several years ago	21 (84)	8 (50)	7 (44)			
Daily frequency of discomfort, *n* (%)				<.001	<0.001	0.165
≤1	12 (48)	1 (6)	0 (0)			
2~3	8 (32)	2 (13)	0 (0)			
4~9	3 (12)	3 (19)	1 (6)			
≥10	2 (8)	10 (60)	15 (94)			
Discomfort severity change, *n* (%)				0.106	<.001	0.109
Getting better	5 (20)	2 (13)	0 (0)			
Same	17 (68)	7 (44)	5 (31)			
Getting worse	3 (12)	7 (44)	11 (69)			
Daily frequency of photopsia, *n* (%)				0.224	0.741	0.377
0	18 (72)	8 (50)	11 (69)			
≤1	4 (16)	1 (6)	2 (13)			
2~3	2 (8)	4 (25)	1 (6)			
4~9	0 (0)	1 (6)	2 (13)			

[a]Kruskal-Wallis nonparametric test and Dunn's Post Hoc Comparisons

Table 6

Statistical analysis of psychological distress across vitreous floater severity levels

	p-values[a]	p-values[b]					
		C - M	C - Mo	C - S	M - Mo	M - S	Mo - S
PSS	0.218	0.318	0.834	0.072	0.383	0.284	0.078
STAI	0.495	0.652	0.362	0.167	0.545	0.237	0.601

[a]Kruskal-Wallis Test nonparametric test
[b]Dunn's Post Hoc Group Comparisons

Table 7

Descriptive statistics of psychological distress across vitreous floater severity groups

	Group	Mean	SD
PSS Score	Control	22.273	6.604
	Mild	23.080	8.592
	Moderate	23.375	4.978
	Severe	26.750	5.745
STAI Score	Control	47.455	11.936
	Mild	48.640	13.988
	Moderate	51.813	12.792
	Severe	53.938	19.884

6. DISCUSSION

PSS and STAI scores between individuals with and without floaters were not statistically different. While the floater group reported higher mean levels of both perceived stress and trait anxiety, these findings do not support a definitive statistical relationship. This result indicates that floater and control participants do not experience different perceived stress and anxiety levels. On the other hand, it may be possible that the insignificant difference may be a result of the limited sample size of both control and floater populations. This result goes against the findings of past research suggesting that older individuals experiencing floaters also experience greater levels of psychological distress (Kim et al., 2017).

Floater participants were divided into three groups: mild, moderate, and severe floater discomfort. Participants reporting more severe floater discomfort were also more likely to report a shorter time since symptom onset. Since individuals experiencing mild floaters were more likely to report a longer time since symptom onset in our study, symptoms may improve over time. This result aligns with previous research on older populations, concluding that floater symptoms were likely to improve within three months if they started within the last month (Park & Choi, 2024). Other studies support this conclusion, suggesting that patients with floaters may adapt over time and come to accept their presence (Cipolletta et al., 2012; Gouliopoulos, 2024). On the other hand, another study finds that the duration of floaters did not improve their utility values and make them adapt to their symptoms (Wagle et al., 2011).

Interestingly, no significant differences were observed regarding the frequency of photopsia and floater severity, which may suggest that photopsia is not associated with psychological distress. This finding is in contrast with a previous study that found photopsias negatively amplified levels of depression and anxiety in patients. This difference may be attributed to populational differences, such as regional and age differences, or limitations of either study (Gouliopoulos, 2025).

Although PSS and STAI scores across floater severity groups were not statistically significant, borderline significant differences in PSS scores were found between the control and severe groups, as well as between the moderate and severe groups. This suggests that greater floater severity may be modestly associated with higher perceived stress levels. This finding is consistent with previous research in older populations that has identified a correlation between floater severity and mental health outcomes such as stress. However, it contrasts with findings from the paper, which reported a positive association between floater severity and anxiety levels (Kim et al., 2017).

7. CONCLUSION

This study aimed to explore the relationship between vitreous floater severity and psychological distress, specifically anxiety and stress levels, in young adults (aged 16 to 18). Individuals experiencing vitreous floaters were not found to have significantly greater psychological distress compared to asymptomatic individuals. However, the degree of floater severity was moderately associated with higher perceived stress levels. Due to the cross-sectional design of this study, causal relationships between these variables cannot be established. Although psychological distress symptoms were not found to be substantial in this younger population, the potential impact on well-being should not be overlooked and support should remain accessible. While vitreous floaters are often dismissed as benign, their subjective effect on individuals may still be meaningful. Clinicians might consider integrating mental health assessments when treating patients with persistent visual disturbances, and patients could explore therapeutic strategies to manage their mental response to floaters.

9. LIMITATIONS

This study contains several limitations that may impact the interpretation of its findings. Firstly, the small sample size, especially the limited number of control participants, limited generalizability, did not account for variability, and reduced the statistical power needed to detect significant differences between groups. Additionally, the use of a cross-sectional design restricts the ability to draw conclusions about causality; while associations were observed, it cannot be determined whether floaters influence psychological distress or vice versa. Despite the non-parametric statistical tests being appropriate for non-normally distributed data, it may reduce the sensitivity of the analyses. Other confounding variables, such as socioeconomic state, mental state, and gender, were not considered, preventing causation to be established between psychological problems and floater symptoms. There are also limitations due to the sample collection method. Because participants got to self-select to participate in this study, people with more extreme opinions were more likely to participate, thereby introducing voluntary response bias. Furthermore, the survey was only distributed on specific Reddit channels, which may over- or underestimate floater participants, not have been proportionally representative of the broader Reddit community, and limit the diversity of responses. This limitation also increases undercoverage bias as people who are not active on the specific Reddit channels were underrepresented in the sample. Another methodological limitation is that, although additional psychological factors were initially intended to be examined, they were excluded in order to keep the survey concise and maintain participant attention. Lastly, this study relied on self-reported measures for both floater severity and psychological symptoms. Participants may underreport or overreport their symptoms due to recall bias, social desirability, or misinterpretation of questions. Floaters and psychological symptoms were not clinically confirmed by an ophthalmologist, which limits the objectivity of symptom classification.

10. FUTURE DIRECTIONS

Future research should aim to collect larger sample sizes of this target population to account for variability and detect smaller differences in the data. Future research should consider a longitudinal approach to observe changes over time and better understand the causal relationship between floater severity and psychological distress. Finally, future research should investigate biomarkers of stress, such as cortisol levels and heart rate variability, alongside self-reports to add objectivity to the stress measurement.

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